Reconstructing Medication Persistence

Transforming fragmented pharmaceutical data into transparent, reproducible metrics



Consolidating claims and enrollment data sources.

Standardizing patient identification criteria.

Implementing quantitative tracking of patient adherence.

Developing interactive dashboards for actionable insights.

Deploying RAG Methodology Assistant for transparency.

Addressing stakeholder challenges by bridging business requirements, field visibility, and analytical rigor.

Problem-Solving Approach and Data Preparation

Standardized Engineering Lifecycle for Data Analytics



Clearly define the project goals and analytical requirements.

Establish inclusion criteria and business logic parameters.

Construct a robust, automated framework for data processing.

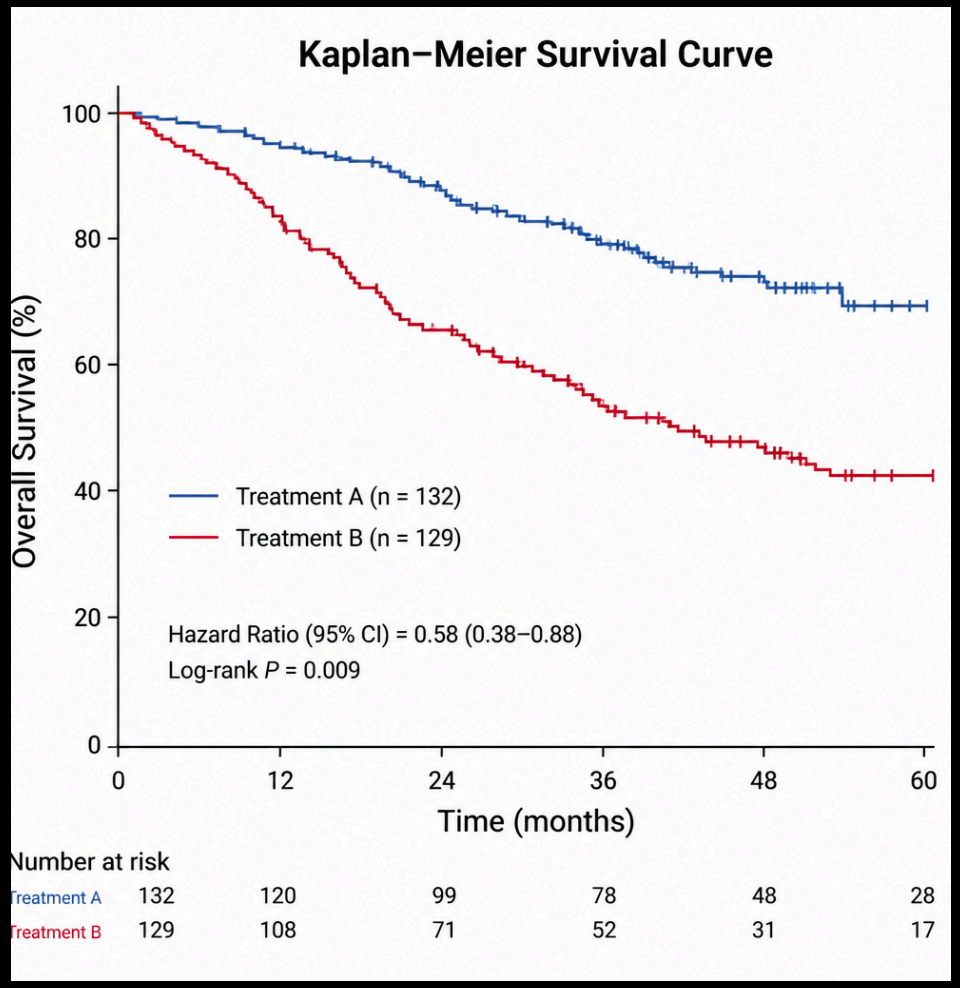
Ensure data integrity through rigorous testing and accuracy checks.

Deploy final solutions for ongoing analysis and reporting.

Pipeline Workflow: Synthetic Data Gen → Validation → Deduplication → Paid Claims Filter → Enrollment Merge. Constraints: Paid claims only, 1-365 days supply filter, and zero use of real patient data.

Cohort Reconstruction and Persistence Methodology

Standardized criteria for health history, treatment initiation, and survival analysis



- **Eligibility Criteria**

  Requires 12-month continuous enrollment for baseline history and strict incident use to capture new treatment initiations.

- **Persistence Definitions**

  Sets a 60-day threshold for allowable gaps and caps refill carry-forward at 30 days to ensure data accuracy.

- **Analytical Comparison**

  Balances a 6-month snapshot for utilization with Kaplan-Meier survival analysis for precise time-to-discontinuation modeling.

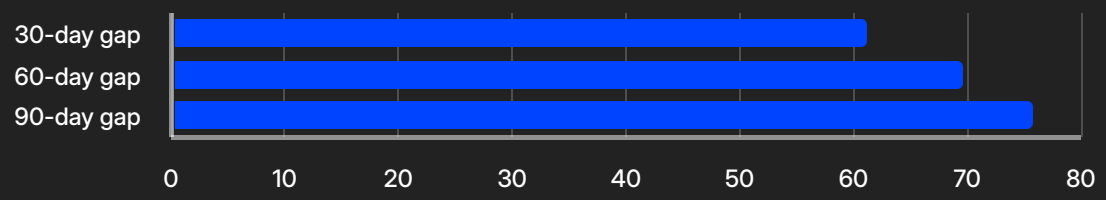
Analytical Results and Sensitivity Analysis

Evaluation of synthetic study outcomes and methodological thresholds

Six-Month Persistence Rate

69.70%

Sensitivity by Refill-Gap Definition



Sample Size

1500

synthetic patients

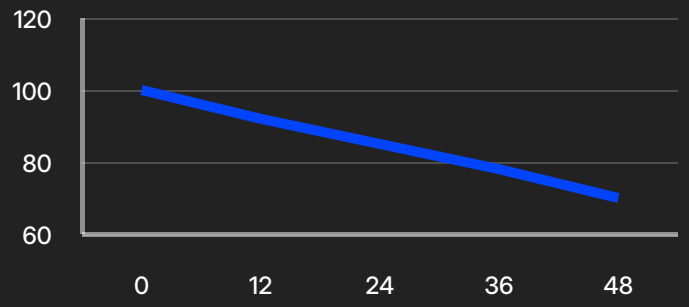
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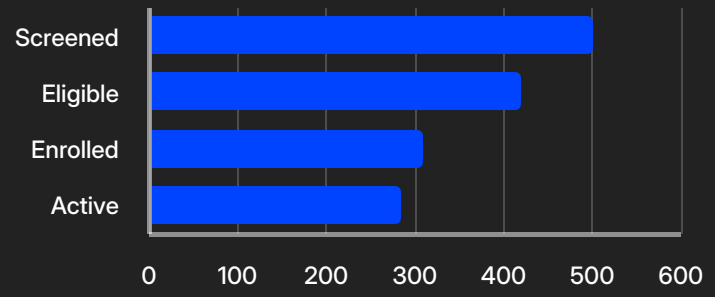
Interactive Dashboard and Technical Architecture

High-performance, local-first clinical data processing

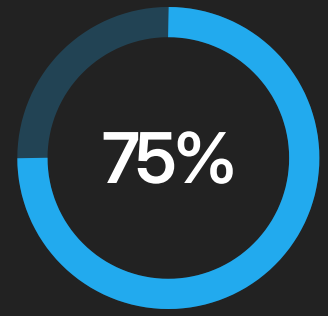
Survival Analysis (Kaplan-Meier)



Cohort Funnel Analysis



Refill-Gap Sensitivity



Data Security

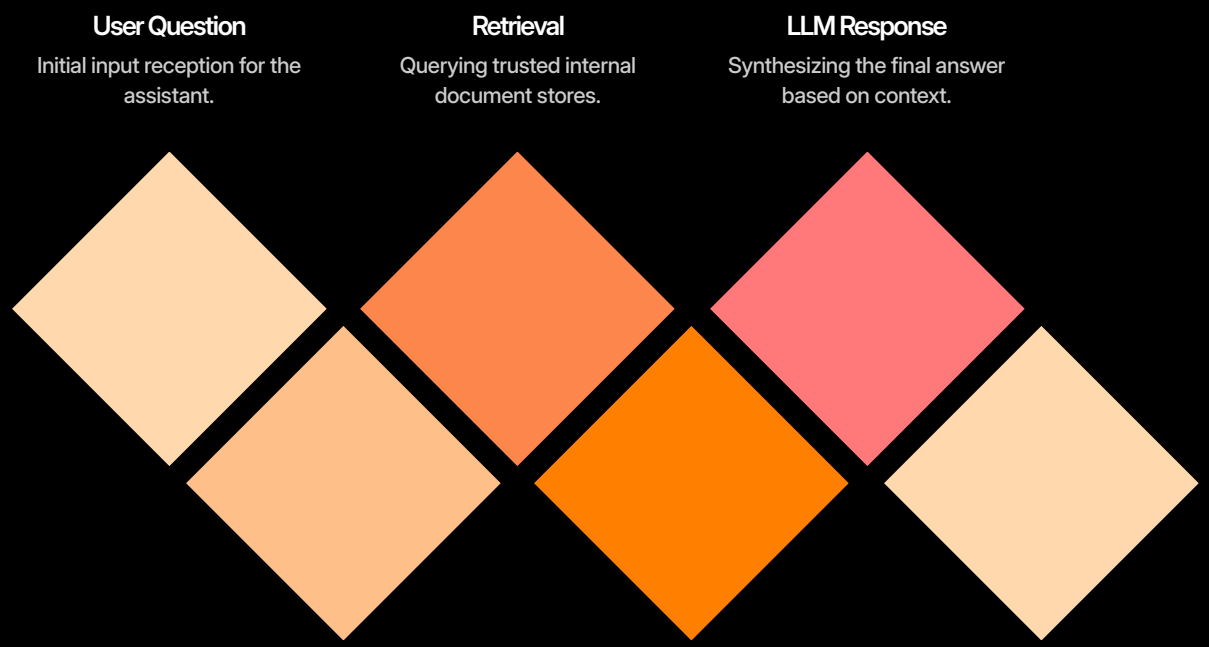
Local-first architecture ensures 0% exposure of patient-level data by eliminating external cloud transit.

Core Tech Stack

Built on Python/Pandas for processing, utilizing native browser rendering (HTML/JS/SVG) for high-performance interactivity.

RAG-Based Methodology Assistant

Transforming static documentation into evidence-grounded AI

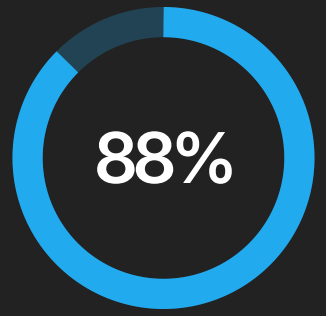


Strict Abstention: The assistant must explicitly abstain from answering if evidence is missing or out-of-scope to prevent hallucination.

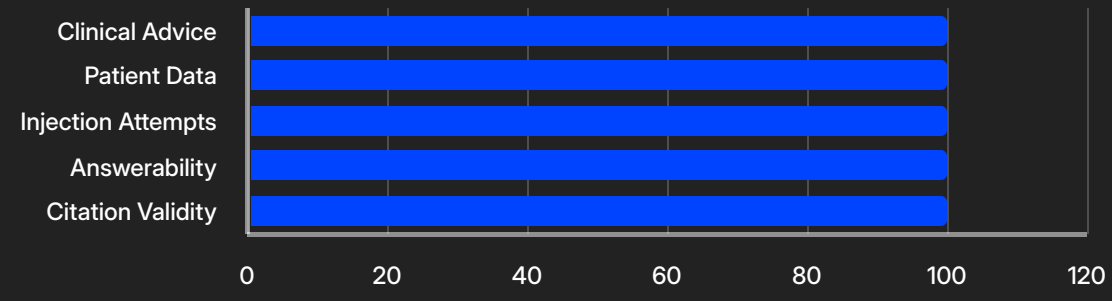
RAG Evaluation: Retrieval & Safety

Performance Metrics and Guardrail Efficacy

Recall@5 (Hybrid + Reranker)



System Performance



Safety Rejection Rate

100%

Success on unsupported queries

Baseline Strategy

Maintain BM25 as the default baseline configuration to ensure long-term system stability and predictable retrieval performance.

Business Value, Limitations, and Next Steps

Standardizing evaluation frameworks for AI reliability and stakeholder trust

Strategic Value

Achieving reproducible metrics, transparent logic, and reduced AI hallucination risks.

Next Steps

Validating methodologies, testing real-world data, and expanding assessment scope.

My goal was not just to calculate a number; it was to build a system stakeholders can trust.